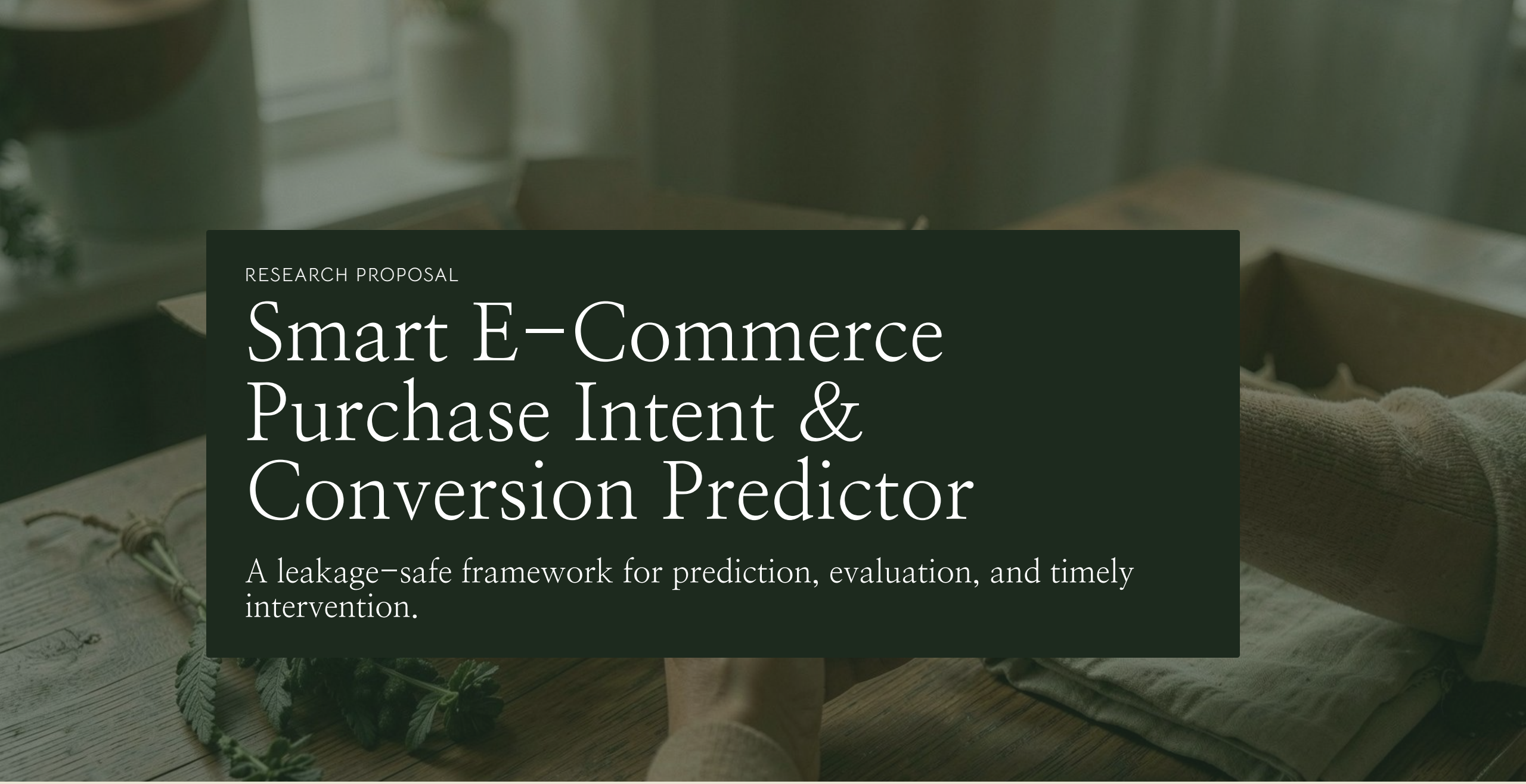


RESEARCH PROPOSAL

Smart E-Commerce Purchase Intent & Conversion Predictor

A leakage-safe framework for prediction, evaluation, and timely intervention.

Made with GAMMA

PROJECT CREDITS

Smart prediction, careful evaluation.

Submitted By

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THE MOTIVATION

Low conversion leaves high-value intent invisible

A large share of sessions ends without a purchase, even when browsing behavior may signal meaningful interest.



Unresolved sessions

E-commerce conversion rates are typically around 15%, leaving many browsing sessions unresolved.



Real-time prediction

The proposal is to predict whether a session is likely to end in purchase.



Earlier action

Use the prediction to identify high-value visitors before they abandon the site.

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SYSTEM LOGIC

The system turns session behavior into an intervention signal

Three stages connect observed browsing behavior to a practical marketing response.

01

Capture

Capture behavioral signals from the browsing session.

02

Estimate

Estimate purchase intent with a trained classification model.

03

Intervene

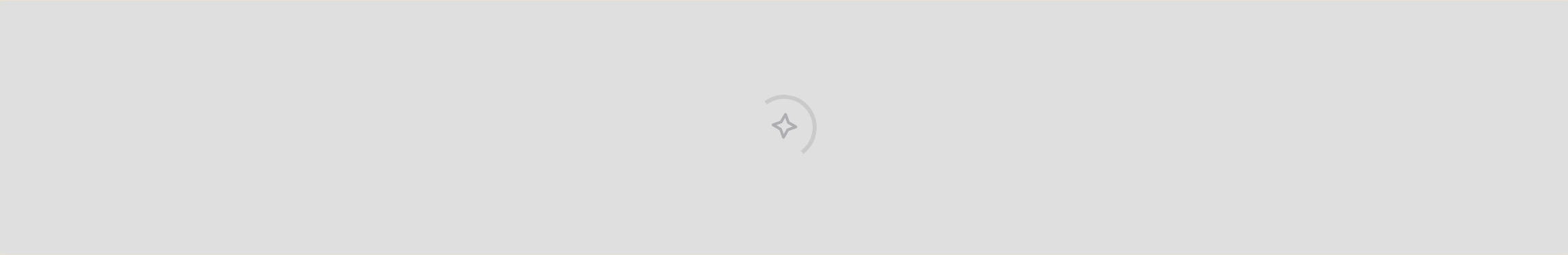
Trigger proactive marketing interventions for high-value visitors.

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EXPERIMENTAL DESIGN

Methodological rigor starts before model training

Leakage control is established in the data-splitting sequence, before any balancing operation is applied.



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MODEL BENCHMARK

Five models create a meaningful comparison

Each candidate is assessed under the same preprocessing and evaluation protocol.

Logistic Regression

Interpretable baseline for purchase classification.

KNN

Tests neighborhood-based decision behavior.

SVM

Tests an alternative separating boundary.

Decision Tree

Tests a rule-based decision structure.

Random Forest

Tests ensemble behavior across multiple trees.

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EVALUATION METRICS

F1-Score and ROC-AUC measure different kinds of success

POSITIVE-CLASS BALANCE

F1-Score

Captures the balance between precision and recall for the positive purchase class.

RANKING QUALITY

ROC-AUC

Measures ranking and discrimination across classification thresholds.

Report both metrics so performance is not reduced to a single number.

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EVALUATION QUESTIONS

The evaluation answers both research and business questions

01

Reliability

Which model identifies likely purchasers most reliably?

02

Comparison

How does performance change across competing algorithms?

03

Usefulness

Can predictions support timely, targeted intervention without overstating certainty?

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PROPOSED CONTRIBUTION

Success means credible prediction that can guide action

Rigor

Deliver a leakage-safe experimental pipeline and comparative model evaluation.

Evidence

Identify the strongest model using F1-Score and ROC-AUC together.

Action

Translate the result into a practical signal for proactive e-commerce marketing.

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